INDEPENDENT STUDY PROPOSAL FORM

Instructions: Please complete all relevant fields and submit.

STUDENT INFORMATION

Student Name	Student E-mail	Student Id
Ma Thm Hng	hang.ma.230142@student.fulbright.edu.vn	230142
ng Thái Hoà	hoa.dang.230165@student.fulbright.edu.vn	230165
Nguyn Trng t	dat.nguyen.240062@student.fulbright.edu.vn	240062

FACULTY INFORMATION

Instructor Name
Dang Huynh

Instructor Status
Full Time

FUV E-mail
dang.huynh@fulbright.edu.vn

COURSE INFORMATION

Course Title
Applied Large Language Models for Recruitment Systems

Vietnamese Course Title
ng dng mô hình ngôn ng ln cho h thng tuyn dng

Academic Year:

Proposed Start Date
07/09/2026

Proposed End Date
24/12/2026

Please indicate if this is a new study or a study based on a previously approved independent study, or a study based on an existing course:

This is a new study.

Please briefly describe the nature of the amendments:

Major/Minor Area:
Note: The form must be approved by major/minor coordinators of all requested majors and minors.

Major/Minor	Course Level
Computer Science	300 (Advanced)

Special Purpose:
In addition to the curriculum purposes determined by the major(s)/minor(s) and course level(s), does this course serve any specific degree requirements, such as a theory course in Major A or an elective applied course in Major B?

No

Course Description:
Please provide a 100-300-word description for the course catalog.

Recruitment processes in Vietnam and globally are increasingly overwhelmed by high CV volumes. Manual screening is time-consuming, inconsistent, and subject to unconscious bias. Existing AI-based ATS (Applicant Tracking Systems) suffer from several limitations: • Poor support for Vietnamese-language CVs due to lack of training data and language-specific tokenization. • Reliance on keyword matching rather than semantic understanding of candidate qualifications. • Opaque scoring systems with no interpretability or explainability for candidates or recruiters. • Dependency on expensive third-party APIs (e.g., OpenAI) that raise data privacy concerns in enterprise contexts. This project addresses these gaps by building an end-to-end system grounded in fine-tuned LLMs.

Prerequisites (please provide course codes):

CS208 - Machine Learning for Data Science

Learning Outcomes:
Please provide a list of the course's learning outcomes using the following format

Upon completion of this course, students will be able to:

- - CLO1. …
- - CLO2. …
- - CLO3. …

*Typically, each course has at least 3 CLOs. Please consider using **Bloom's taxonomy** when writing these CLOs.*

- CLO1: Analyze fairness, robustness, and alignment challenges in AI powered recruitment and CV screening systems.
- CLO2: Develop and evaluate a fine-tuned large language model for multilingual CV parsing and job fit assessment.
- CLO3: Design, document, and present an interpretable and reproducible AI system for real world deployment.

Mapping CLO and MLO Charts:
Please match the course's learning outcomes to the major/minor's learning outcomes using the table format provided in the textbox below. The list of the major/minor learning outcomes should be included at the end for reference. For the most updated major/minor learning outcomes please see **Major/Minor Descriptions and Flowcharts.**

A sample mapping is as follows:

	MLO1	MLO2	MLO3	MLO4	MLO5	MLO6	MLO7	MLO8
CLO1	X	X	X			X		
CLO2	X		X	X	X			
CLO3	X	X		X	X	X	X	
CLO4	X	X		X	X	X	X	X

This table can be copied, pasted in the textbox, and modified as needed.

Note: if the course is cross-listed, its learning outcomes must be mapped to the learning outcomes of the cross-listed majors and minors as well (one additional mapping table for every cross-listed major/minor).

Mapping Major/Minor 1:

	MLO1	MLO2	MLO3
CLO1	x	x	
CLO2	x	x	x
CLO3	x		x

Computer Science MLOs:
MLO1: Demonstrate understanding of fundamental CS by illustrating core concepts, comparing methods, and solving problems in theory and systems.
MLO2: Explain emerging aspects of the CS discipline (e.g. AI, machine learning, data science)
MLO3: Develop real-world applications using knowledge of CS and other disciplines

Rationale:
Please provide a brief explanation of how the proposed course supports the intended major/minor area and, if applicable, other major/minor areas in the undergraduate program. Please also comment on how this course might evolve in the future and how it can be sustainable.

As AI recruitment tools become more widespread, building systems that are accurate, fair, and explainable is increasingly important. This project addresses that challenge by fine tuning an open source LLM on Vietnamese and English CV data without relying on third party APIs, thereby ensuring data privacy and full control over model behavior. The project includes multilingual CV parsing, job fit evaluation, and interpretable scoring mechanisms within an end to end recruitment support system. The result is a production ready web application that demonstrates the practical application of large language models in real world recruitment and candidate evaluation tasks.

Tentative Syllabus:
Please provide the main topics, textbooks, and required readings for the course.

Textbook:
Natural Language Processing with Transformers - Lewis Tunstall et al. (O'Reilly, 2022)
Build a Large Language Model From Scratch - Sebastian Raschka (2024)
Speech and Language Processing - Jurafsky & Martin (3rd ed.)
Applied Text Analysis with Python - Bengfort et al. (O'Reilly)

tentative weekly topics:

weeks 1–3: foundations and data preparation
- review transformers, deep learning, and ai alignment concepts
- survey recruitment ai systems and cv parsing methods
- collect and preprocess vietnamese and english cv–jd datasets

weeks 4–7: model development and evaluation
- set up and fine tune llms using lora/qlora
- develop cv extraction and job fit scoring modules
- evaluate model performance using quantitative and human evaluation

weeks 8–11: system development and analysis
- develop backend apis and frontend interface
- integrate interpretability and explanation modules
- conduct adversarial testing and bias analysis

weeks 12–15: reporting and deployment
- finalize codebase and technical documentation
- publish models and source code on github and hugging face
- prepare final report, presentation, and live demonstration

Assessment Types:
Please list the assessment types only (e.g., exams, essays, problem sets, presentations). Detailed explanations should be provided in the subsequent field.

- Technical report - GitHub repository - Final Project Presentation

Assessments:
For each assessment type, please indicate the associated weight and how it measures the course learning outcomes (CLOs). If the assessment type is non-typical for the course level, please briefly explain why it is appropriate.

- - technical report (30%): a 1000–1500 word report describing the research question, methodology, experiments, and findings. assessed based on clarity, rigor, critical thinking, and alignment analysis.
  - github repository (60%): a clean and modular codebase including scripts, logs, and readme documentation to ensure reproducibility.
  - final project presentation (10%): a final presentation with a live demonstration and deployment of the application, along with public release of the report and code repository.

clos mapping:

- Technical report (30%): clo1,clo2,clo3
- GitHub repository (60%): clo1,clo2,clo3
- Final Project Presentation (10%): clo2,clo3

Written Coursework:
For courses with written assignments, explain how writing pedagogy (appropriate to the course level) will be incorporated (e.g., student peer review, writing workshops, review of paper proposals, review of drafts, in-class writing instruction).
For courses in which most assessments are written, please be aware of the recommended wordcount limits:
100-level: approx. 3000 words across all assignments.
200-level: approx. 3500 words across all assignments.
300-level: approx. 4000 words count across all assignments.

Meeting Plan:
Please provide the frequency and format of meetings.

Weekly 2-hour in-person meetings

ADDITIONAL INFORMATION

Field Trip(s):
If yes, please briefly explain the nature of the field trip(s)and provide details regarding proposed scheduling, funding, and sustainability.

No

Lab Component(s):
If yes, please briefly explain the nature of the lab component(s)and provide details regarding proposed scheduling, funding, and sustainability.

No

Facilities and Resources:
If the course requires special facilities or equipment resources, please explain here.
(Note: filmmaking equipment is available for select borrowing and priority booking by the instructor for designated practicum courses. Please review the Equipment Loan Policy for additional details.)

Commitment(s) Beyond Normal Course Hours:
A 4-credit course needs to follow the academic calendar and scheduled timetable for a total of 42 hours of teaching hours and should not exceed those hours.
If applicable, please provide a brief description of the activities (movie screenings, special lectures, etc.), an estimate of the additional hours beyond the normally scheduled class time required for these activities, and a justification.
Additionally, please provide a tentative schedule for the additional hours/commitments (weekday evenings, weekends, etc.).

Disability and Access:
Does this course include any aspects which might affect its accessibility to a diverse community of students? If so, what actions are needed (from faculty, Academic Affairs, Learning & Teaching, Facilities, or any other part of the university) to support equality of access?

Additional Comments or Requests: